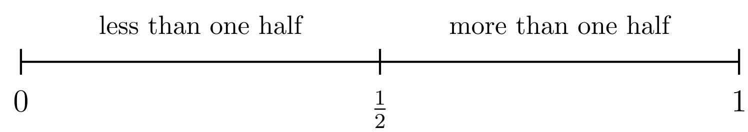
Comparing Unlike Fractions in Real Situations

Grade 4–5 Mathematics

Benchmarks, common denominators, equivalent fractions, and ordering. Every problem below gives you two or more fractions of the *same whole* — the same size bottle, the same trail, the same measuring cup. Decide which is larger by the quickest honest method: a benchmark (0, $\frac{1}{2}$, or 1) when the fractions land on opposite sides of it, and a common denominator when they do not. Show the thinking you used, then write your comparison on the answer line.



Reference for the whole sheet: the three benchmarks. No problem's numbers are shown here — use the numbers given in each problem.

1. Ana and Ben each carry the same size water bottle. Ana's bottle is $\frac{2}{5}$ full. Ben's bottle is $\frac{3}{4}$ full. Compare each fraction to the benchmark $\frac{1}{2}$, then write $<$ or $>$ between them: $\frac{2}{5}$ _____ $\frac{3}{4}$.

Answer: ________________

2. A tray of cornbread is cut into 12 equal pieces. Dev says he took $\frac{3}{4}$ of the tray. Write the fraction of the tray Dev took using twelfths, so it can be counted in pieces.

Answer: ________________

3. On Saturday, Sam walked $\frac{2}{3}$ of a mile and Kim walked $\frac{5}{8}$ of a mile. Both fractions are close to $\frac{1}{2}$, so rewrite them with the common denominator 24 and compare: $\frac{2}{3}$ _____ $\frac{5}{8}$.

Answer: ________________

4. Two crews are painting the same fence. The morning crew painted $\frac{4}{9}$ of it and the afternoon crew painted $\frac{5}{6}$ of it. Use the benchmark $\frac{1}{2}$ to compare: $\frac{4}{9}$ _____ $\frac{5}{6}$.

Answer: ________________

5. A recipe card measures everything in fortieths of a cup. The recipe calls for $\frac{7}{10}$ of a cup of oats. Write that amount as an equivalent fraction with denominator 40.

Answer: ________________

6. Mia has read $\frac{5}{6}$ of her book and Jonah has read $\frac{7}{9}$ of the same book. Rewrite both with a common denominator, then compare: $\frac{5}{6}$ _____ $\frac{7}{9}$.

Answer: __________________

7. Three families each planted part of a community garden bed of the same size: the Ortiz family planted $\frac{3}{5}$ of a bed, the Baker family planted $\frac{1}{2}$ of a bed, and the Chen family planted $\frac{7}{10}$ of a bed. List the three fractions in order from least to greatest.

Answer: __________________

8. Two nearly full gas tanks hold the same amount when full. One is $\frac{7}{8}$ full and the other is $\frac{9}{10}$ full. Neither benchmark $\frac{1}{2}$ nor rounding to 1 settles it, so compare how far each one is *from* a full tank: $\frac{7}{8}$ _____ $\frac{9}{10}$.

Answer: __________________

9. Two granola recipes make the same size batch. Recipe A uses $\frac{11}{15}$ of a cup of honey and Recipe B uses $\frac{3}{4}$ of a cup. Both are a little more than $\frac{1}{2}$, so a benchmark will not decide it. Find a common denominator and compare: $\frac{11}{15}$ _____ $\frac{3}{4}$.

Answer: __________________

10. Four hikers report how much of the same trail they finished before lunch: Rosa $\frac{7}{12}$, Tam $\frac{5}{8}$, Wes $\frac{2}{3}$, and Nia $\frac{3}{4}$. Order the four fractions from greatest to least, and name the hiker who walked the farthest.

Answer: __________________